Polynomial Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 4$ in graphs and characteristics of polynomial functions, evaluate $f(5)$.

Final Answer: 24. Substitute $x = 5$: $f(5) = 4(5) + 4 = 24$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 5$ in solving polynomial equations, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 5(2) + 5 = 15$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for solving polynomial equations consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 6$ in remainder and factor theorems, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 6(2) + 6 = 18$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for remainder and factor theorems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 7$ in roots and zeros, evaluate $f(2)$.

Final Answer: 11. Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for roots and zeros consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Polynomial Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Show that $(x - 2)$ is a factor of $p(x) = x^3 - 3x^2 + 4$, then fully factorise $p(x)$.

$p(2) = 8 - 12 + 4 = 0$ ✓. Divide: $p(x) = (x-2)(x^2 - x - 2) = (x-2)(x-2)(x+1) = (x-2)^2(x+1)$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Trigonometric Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

If $f(x) = x^2 - 1$ and $g(x) = \sqrt{x + 2}$, find (a) $f(g(3))$, (b) the domain of $g \circ f$.

(a) $g(3) = 5$; $f(5) = 5 - 1 = 4$. (b) $g(f(x)) = \sqrt{x^2 - 1 + 2} = \sqrt{x^2 + 1}$; always real, domain: all reals.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 1 · FoundationsLet $q(x) = -(x + 2)^2(x - 1)(x - 4)^3$. What is the degree of $q(x)$, does the graph cross or touch the x-axis at $x = -2$, and what is $\lim_{x \to \infty} q(x)$?

Sum the exponents of the linear factors to find the degree: $2 + 1 + 3 = 6 \Rightarrow \text{degree } 6$. At $x = -2$, the multiplicity is 2 (even), so the graph touches the x-axis and turns around without crossing. The degree is even (6) and the leading coefficient is negative (-1), so $\lim_{x \to \infty} q(x) = -\infty$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Radicals & Rational Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Rational Expressions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 8.01 Conic Sections**

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 9.01 Trigonometric Functions**

Find the coefficient of x^2 in the expansion of $(3 - 2x)^5$.

$\binom{5}{2}(3)^3(-2x)^2 = 10 \times 27 \times 4x^2 = 1080x^2$. Coefficient: 1080.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 1.01 Functions & Their Graphs**

Find the value(s) of k for which $A = (k, 2; 3, k)$ is singular.

$A = k^2 - 6 = 0k = 6$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 10.01 Matrices & Systems**

Level 3 · AdvancedLet $R(x) = \frac{2x^2 - 8}{x^2 - 5x + 6}$. (a) Find the domain of $R(x)$. (b) Identify any holes and state their coordinates. (c) Write the equations of all vertical and horizontal asymptotes.

(a) Domain: Factor the denominator: $x^2 - 5x + 6 = (x - 2)(x - 3)$. Thus the domain is all real numbers except $xeq2$ and $xeq3$, or $(-\infty, 2) \cup (2, 3) \cup (3, \infty)$. (b) Holes: Factor the numerator: $2(x^2 - 4) = 2(x - 2)(x + 2)$. The factor $(x - 2)$ cancels: $R(x) = \frac{2(x+2)}{x-3} \quad (xeq2)$. Evaluating at $x = 2$: $y = \frac{2(2+2)}{2-3} = \frac{8}{-1} = -8$. Hole coordinates: $(2, -8)$. (c) Asymptotes: Vertical asymptote at the remaining denominator zero: $x = 3$. Degrees of nu.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 11.01 Probability & Statistics**

Level 2 · MediumSolve the logarithmic equation $\log_2(x + 3) + \log_2(x - 1) = 5$ analytically for all real solutions. Show the verification step that eliminates extraneous roots.

Step 1 (Apply Product Property): $\log_2((x + 3)(x - 1)) = 5$. Step 2 (Convert to Exponential Form): $(x + 3)(x - 1) = 2^5 = 32$. Step 3 (Solve the Quadratic Equation): $x^2 + 2x - 3 = 32 \Rightarrow x^2 + 2x - 35 = 0$. $(x + 7)(x - 5) = 0 \Rightarrow x = -7 \quad \text{or} \quad x = 5$. Step 4 (Domain Verification): For $x = -7$: $x - 1 = -8 \leq 0$, and $\log_2(-8)$ is undefined in the real numbers. Hence $x = -7$ is an extraneous solution. For $x = 5$: $5 + 3 = 8 > 0$ and $5 - 1 = 4 > 0$. Checking.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Level 2 · MediumThe function $f(x) = x^3 - 3x^2 - 9x + 2$ has a local maximum and a local minimum. Determine the intervals on which f is concave up and concave down, and find the coordinates of the inflection point.

The rate of change switches from decreasing to increasing at the inflection point. Taking the second derivative (or analyzing symmetric rates of change): $f'(x) = 3x^2 - 6x - 9, \quad f''(x) = 6x - 6 = 0 \Rightarrow x = 1.$ For x. For $x > 1$, $f''(x) > 0$, so f is concave up on $(1, \infty)$. The inflection point is $(1, f(1)) = (1, 1 - 3 - 9 + 2) = \mathbf{(1, -9)}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 9$ in review: operations with polynomials, evaluate $f(2)$.

a. 13 ✓ **b.** 15

c. 4 **d.** 17

Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 3$ in synthetic and long division, evaluate $f(2)$.

a. 9 ✓ **b.** 12

c. 6 **d.** 13

Substitute $x = 2$: $f(2) = 3(2) + 3 = 9$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: operations with polynomials consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5 **b.** 17.5

c. 18.5 ✓ **d.** 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for synthetic and long division consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The Intermediate Value Theorem states that if $P(a)$ and $P(b)$ have opposite signs, then:

a. There are two roots

b. **There is a root between a and b** ✓

c. $P(0) = 0$

d. $P(a)P(b) = 0$

IVT guarantees at least one root in (a, b).

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?

a. 24 ✓

b. 18

c. 17

d. 14

We solve the problem in two key phases: identifying the polynomial $f(x)$ from its graphical features, and applying the Polynomial Remainder Theorem to determine $R(x)$. **Step 1: Determine $f(x)$** - Because the graph is tangent to the x-axis at $x = -2$, $(x + 2)$ is a factor with an even multiplicity of at least 2. - Because the graph crosses the x-axis at $x = 1$ and $x = 3$, $(x - 1)$ and $(x - 3)$ are factors with odd multiplicities of at least 1. - Since $f(x)$ is of degree 4, the sum of multipliciti.

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?

a. $f(x) = (x+2)(x-1)(x-3)$ ✓

b. $f(x) = (x-2)(x+1)(x+3)$

c. $f(x) = -(x+2)(x-1)(x-3)$

d. $f(x) = (x+2)(x-1)^2(x-3)$

To determine which polynomial function matches the graph, we analyze the intercepts, multiplicities, and end behavior: 1. **Identify the x-intercepts (zeros):** The graph crosses the x-axis at $x = -2$, $x = 1$, and $x = 3$. By the Factor Theorem, if c is a zero of $f(x)$, then $(x - c)$ is a factor. Therefore, the corresponding linear factors are: $(x - (-2)) = (x + 2)$, $(x - 1)$, and $(x - 3)$ 2. **Multiplicity:** At each of the three x -intercepts, the graph passes directly through.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which of the following statements describes the end behavior of the polynomial function $f(x) = -5x^4 + 2x^3 - 7x + 3$?

a. **As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to -\infty$** ✓

b. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to -\infty$

c. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to \infty$

d. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to \infty$

To determine the end behavior of a polynomial function, examine its leading term, which dominates the function's values for very large positive and negative values of x. 1. Identify the leading term: The highest power of x is 4, so the leading term is $-5x^4$. 2. Analyze the degree and leading coefficient: - The degree is $n = 4$, which is an even integer. - The leading coefficient is $a = -5$, which is negative. 3. Determine the limits as $x \to \pm\infty$: - As $x \to \infty$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Let $P(x)$ be a monic quartic polynomial with real coefficients whose graph is tangent to the x-axis at $x = -3$ and $x = 7$. A horizontal line $y = k$ intersects the graph of $y = P(x)$ at four distinct points whose x-coordinates, when listed in increasing order, form an arithmetic progression. What is the value of k?

a. 400 ✓

b. 225

c. 256

d. 625

1. **Determine $P(x)$:** Because $P(x)$ is a monic quartic polynomial that is tangent to the x-axis at $x = -3$ and $x = 7$, it has roots of multiplicity at least 2 at both points. Since $\deg(P) = 4$, these multiplicities must be exactly 2: $P(x) = (x + 3)^2(x - 7)^2$ 2. **Exploit the axis of symmetry:** The line of symmetry lies halfway between -3 and 7: $h = \frac{-3+7}{2} = 2$ Let $u = x - 2$. Then $x + 3 = u + 5$ and $x - 7 = u - 5$. In terms of u, the polynomial becomes.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following is a polynomial function?

a. $f(x) = \frac{1}{x}$

b. $g(x) = \sqrt{x}$

c. $h(x) = 3x^4 - 2x^2 + 5x - 1$ ✓

d. $k(x) = 2^x$

A polynomial function is a function of the form $P(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$, where a_i are constants and n is a non-negative integer. Option 1, $f(x) = \frac{1}{x}$, can be written as x^{-1}, which has a negative exponent, so it is not a polynomial. Option 2, $g(x) = \sqrt{x}$, can be written as $x^{1/2}$, which has a fractional exponent, so it is not a polynomial. Option 3, $h(x) = 3x^4 - 2x^2 + 5x - 1$, has all terms with non-negative integer exponents (4, 2, 1, and 0 for the co.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Show that the sum of zeros of a monic polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ equals $-a_{n-1}$.

a. Cannot be determined

b. **By Vieta: expanding $\prod(x - r_i)$, the x^{n-1} coefficient is $-\sum r_i = a_{n-1}$** ✓

c. Sum equals a_0

d. Sum equals n

Applying Polynomial Functions principles, Vieta's formulas. Therefore, the verified correct option is By Vieta: expanding $\prod(x - r_i)$, the x^{n-1} coefficient is $-\sum r_i = a_{n-1}$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Use the IVT to show $P(x) = x^3 - 2$ has a root between 1 and 2.

a. IVT does not apply

b. $P(1) = -10$**, so by IVT a root exists in** $(1, 2)$ ✓

c. $P(0) = 0$

d. $P(1) > 0$ and $P(2) > 0$, no root

Applying Polynomial Functions principles, Sign change implies a root. Therefore, the verified correct option is $P(1) = -10$, so by IVT a root exists in $(1, 2)$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the cubic polynomial $P(x)$ with leading coefficient 2 whose graph passes through $(0, 6)$ and has zeros $1, -3$ and $1 + i$. But wait — to have real coefficients, if $1 + i$ is a root, so is $1 - i$. That gives degree 4.

a. $P(x) = 2x^3 - 6$

b. $P(x) = 2(x - 1)(x + 3)(x - 1 - i)$

c. $P(x) = 2(x - 1)^2(x + 3)$

d. **This is impossible for a cubic with real coefficients (need degree ≥4 for those zeros)** ✓

Complex roots of polynomials with real coefficients come in conjugate pairs, so $1 + i$ forces $1 - i$ as well → at least degree 4. Impossible for cubic.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use synthetic division to show $(x + 1)$ divides $x^{100} - 1$.

a. Only for odd powers

b. $P(-1) = (-1)^{100} - 1 = 1 - 1 = 0$**. Yes.** ✓

c. $P(-1) \neq 0$

d. Indeterminate

Applying Polynomial Functions principles, Even power: $(-1)^{100} = 1$. Therefore, the verified correct option is $P(-1) = (-1)^{100} - 1 = 1 - 1 = 0$. Yes..

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Use the Rational Root Theorem: list possible rational roots of $2x^3 - 3x + 1 = 0$.

a. $\pm 3,\ \pm \frac{1}{2}$

b. $\pm 1,\ \pm \frac{1}{2}$ ✓

c. ± 1

d. $\pm 1,\ \pm 2$

Applying Polynomial Functions principles, $\pm \frac{p}{q}$: $p \mid 1$, $q \mid 2$. Therefore, the verified correct option is $\pm 1,\ \pm \frac{1}{2}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Write a cubic with roots $1, 2, 3$.

a. $x^3 - 6x^2 - 11x + 6$

b. $x^3 + 6x^2 + 11x + 6$

c. $(x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6$ ✓

d. $(x+1)(x+2)(x+3)$

Applying Polynomial Functions principles, Product of linear factors. Therefore, the verified correct option is $(x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Apply Descartes' Rule of Signs to $P(x) = x^5 + x^4 - 3x^3 - 3x^2 + 2x + 2$.

a. At most 4 positive real roots

b. At most 1 positive

c. **At most 2 positive real roots** ✓

d. Exactly 3 positive, exactly 2 negative

Sign sequence $+, +, -, -, +, +$: 2 sign changes $\rightarrow \leq 2$ positive. $P(-x)$: $-x^5 + x^4 + 3x^3 - 3x^2 - 2x + 2 \rightarrow$ signs $-, +, +, -, -, +$: 3 changes $\rightarrow \leq 3$ negative.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Matrices & Systems

The polynomial $P(x)$ has degree 4 and zeros $1, 1, 2, -3$. Write it in factored form (monic).

a. $(x+1)^2(x-2)(x+3)$

b. $(x-1)^2(x-2)(x+3)$ ✓

c. $(x-1)(x-2)(x+3)$

d. $(x-1)^2(x+2)(x-3)$

Applying Polynomial Functions principles, Multiplicity 2 at $x = 1$. Therefore, the verified correct option is $(x-1)^2(x-2)(x+3)$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability & Statistics

Find a degree-3 polynomial with zeros $0, -1, 4$ and leading coefficient 2.

a. $x(x-1)(x+4)$

b. $2x(x+1)(x-4) = 2x^3 - 6x^2 - 8x$ ✓

c. $2x^2(x-4)$

d. $2x(x-1)(x+4)$

Applying Polynomial Functions principles, $2x(x+1)(x-4)$. Therefore, the verified correct option is $2x(x+1)(x-4) = 2x^3 - 6x^2 - 8x$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the quotient when $x^4 - 1$ is divided by $(x-1)$.

a. $x^3 + x^2 + x + 1$ ✓

b. $x^3 + 1$

c. $x^4 + 1$

d. $x^3 - x^2 + x - 1$

Applying Polynomial Functions principles, $(x-1)(x^3 + x^2 + x + 1) = x^4 - 1$. Therefore, the verified correct option is $x^3 + x^2 + x + 1$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rational Expressions

Find the value of k such that $(x-3)$ is a factor of $P(x) = x^3 - kx^2 + 2x - 3$.

a. $k = \frac{10}{9}$

b. $k = 3$

c. $k = 2$

d. $k = \frac{10}{3}$ ✓

By the Factor Theorem, $P(3) = 0 \Rightarrow 3^3 - k(3)^2 + 2(3) - 3 = 0 \Rightarrow 30 - 9k = 0 \Rightarrow k = \frac{10}{3}$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Determine: is $(x + 2)$ a factor of $P(x) = x^3 + x^2 - 4x - 4$?

a. Only if degree is odd

b. No

c. **Yes,** $P(-2) = 0$ ✓

d. Cannot determine

Applying Polynomial Functions principles, $P(-2) = -8 + 4 + 8 - 4 = 0$. Yes. Therefore, the verified correct option is Yes, $P(-2) = 0$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the complex roots of $x^4 - 1 = 0$.

a. $x = \pm i$

b. $x = 1, -1, i, -i$ ✓

c. $x = 1, i$

d. $x = \pm 1$

Applying Polynomial Functions principles, $x^4 - 1 = (x^2 - 1)(x^2 + 1)$. Therefore, the verified correct option is $x = 1, -1, i, -i$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

How many turning points can a degree-n polynomial have (maximum)?

a. $n + 1$

b. $n - 1$ ✓

c. n

d. $2n$

Applying Polynomial Functions principles, At most $n - 1$ turning points. Therefore, the verified correct option is $n - 1$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Conic Sections

Find all real zeros of $f(x) = x(x - 2)(x + 3)$.

a. $x = 0, -2, 3$

b. $x = 0, 2, 3$

c. $x = 0, 2, -3$ ✓

d. $x = 2, -3$

Applying Polynomial Functions principles, Set each factor to 0. Therefore, the verified correct option is $x = 0, 2, -3$.

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